Math 421/510: Homework 1

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1. Define $d_1(x, y) = |\arctan(x) - \arctan(y)|$ for $x, y \in \mathbb{R}$. Show that:

   (a) d_1 is a metric on $\mathbb{R}$;

   (b) $(\mathbb{R}, d_1)$ is not complete;

   (c) d_1 produces the same topology (collection of open sets) on $\mathbb{R}$ as the usual metric $d(x, y) = |x - y|$.

Solution.

(a) We show that $d_1(x, y) = 0 \implies x = y$, $d_1(x, y) = d_1(y, x) \quad \forall x, y \in \mathbb{R}$, and the triangle inequality.

- $d_1(x, y) = 0$ implies $x = y$:

$$0 = d_1(x, y) = |\arctan(x) - \arctan(y)| \implies \arctan(x) = \arctan(y).$$

  Since $\arctan(x)$ is bijective, this implies $x = y$.

- Symmetry:

$$\begin{aligned} d_1(x, y) = |\arctan(x) - \arctan(y)| &= |-(\arctan(y) - \arctan(x))| \\ &= |\arctan(y) - \arctan(x)| = d_1(y, x). \end{aligned}$$

- Triangle Inequality:

$$\begin{aligned} d_1(x, z) = |\arctan(x) - \arctan(z)| &= |\arctan(x) - \arctan(y) + \arctan(y) - \arctan(z)| \\ &\leq |\arctan(x) - \arctan(y)| + |\arctan(y) - \arctan(z)| \\ &= d_1(x, y) + d_1(y, z). \end{aligned}$$

(b) Consider the sequence $\{x_n = n\}_{n \in \mathbb{N}}$. We show that this sequence is cauchy in $(\mathbb{R}, d_1)$. Let $\epsilon > 0$. Then, $\lim_{n \to \infty} \arctan(x_n) = \pi/2$ (one of the properties of $\arctan(x)$) so there exists some N such that for all $n > N$, $|\pi/2 - \arctan(x)| < \epsilon/2$ (since $\arctan(x)$ is strictly increasing). But then, for any $n, m > N$, we have

$$d_1(x_n, x_m) = |\arctan(x_n) - \arctan(x_m)| \leq |\arctan(x_n) - \pi/2| + |\pi/2 - \arctan(x_m)| \leq \epsilon/2 + \epsilon/2.$$

Now, suppose that $\lim_{n \to \infty} d_1(x_n, z) = 0$ for some $z \in \mathbb{R}$. Then, there exist some N such that $z < N$. However, then we have that $d_1(z, x_n) > d_1(z, x_N) > 0$ for all $n > N$, which contradicts that $\lim_{n \to \infty} d_1(x_n, z) = 0$. Thus, this sequence has no limit in $(\mathbb{R}, d_1)$.

(c) Let $B_d(x, \delta) = \{y \in \mathbb{R} : d(x, y) < \delta\}$. By the definition of open sets, d and d_1 produce the same open sets iff for any $B_d(x, \delta)$ there exists δ' such that $B_{d_1}(x, \delta') \subset B_d(x, \delta)$ and vice versa.

Let $x \in \mathbb{R}$, $\delta > 0$. Then, there exists some $y \in (-\pi/2, \pi/2)$ such that $\arctan(x) = y$. But then, $\tan(y) = x$, and since $\tan(x)$ is continuous at x, we have that there exists some $\delta' > 0$ such that for all $|y - z'| < \delta'$, $|\tan(y) - \tan(z')| < \delta$. However, this implies that for

all z with $|\arctan(x) - \arctan(z)| < \delta'$, $|\tan(\arctan(x)) - \tan(\arctan(z))| = |x - z| < \delta$. Therefore, $B_{d_1}(x, \delta') \subset B_d(x, \delta)$.
For the other direction, let $x \in \mathbb{R}, \delta > 0$. Now, since $\arctan'(x) = \frac{1}{1+x^2}$, we have that $0 < \arctan'(x) \leq 1$ for all x. Then, by the mean value theorem, this implies $|\arctan(x) - \arctan(y)| = \arctan'(z) \cdot |x - y|$ for some $z \in (x, y)$, so $|\arctan(x) - \arctan(y)| \leq 1 \cdot |x - y|$. Therefore, if $|x - y| < \delta$, then $|\arctan(x) - \arctan(y)| < \delta$, so $B_d(x, \delta) \subset B_{d_1}(x, \delta)$, as desired.

2. Consider the following spaces of sequences:

$$\ell^\infty = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \|x\|_\infty = \sup_j |x_j| < \infty\} \qquad \text{(bounded sequences)}$$

$$\ell_0 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \lim_{j \to \infty} x_j = 0\} \qquad \text{(sequences tending to 0)}$$

$$\ell_c = \{x = (x_1, x_2, \dots, x_m, 0, 0, \dots) \mid x_j \in \mathbb{C},\ m \in \mathbb{N}\} \qquad \text{(finite sequences).}$$

Note that: $\ell_c \subset \ell_0 \subset \ell^\infty$, and all are vector spaces.

(a) show that $\|\cdot\|_\infty$ is a norm on ℓ^∞ (hence also on ℓ_c and ℓ_0);

(b) in class we showed that $(\ell^\infty, \|\cdot\|_\infty)$ is complete (hence Banach). Show that ℓ_c is not complete in the metric produced by $\|\cdot\|_\infty$;

(c) show that $(\ell_0, \|\cdot\|_\infty)$ is complete (hence Banach);

(d) show that ℓ_0 is the closure of ℓ_c (the smallest closed set which contains it) in $(\ell^\infty, \|\cdot\|_\infty)$.

(a) Suppose that $z \in \ell^\infty$ is such that $\|z\|_\infty = 0$. In particular, $\sup_j |(z)_j| = 0$. However, this implies $|(z)_j| \leq 0$, so $z_j = 0$. Thus, $z = 0$.
Let $z \in \ell^\infty$, and let $\lambda \in \mathbb{C}$. Now, for each index j, we have $(\lambda z)_j = \lambda(z)_j$. Then,

$$\|\lambda z\|_\infty = \sup_j |(\lambda z)_j| = \sup_j |\lambda(z)_j| = \sup_j |\lambda||(z)_j| = |\lambda| \sup_j |(z)_j| = |\lambda|\|z\|_\infty.$$

Finally, we prove the triangle inequality. Let $y, z \in \ell^\infty$ We have that:

$$\|y + z\| = \sup_j |(y + z)_j| \leq \sup_j (|(y)_j| + |(z)_j|) \leq \sup_j |(y)_j| + \sup_k |(z)_k| = \|y\|_\infty + \|z\|_\infty.$$

Therefore, $\|\cdot\|_\infty$ is a norm over ℓ^∞.

(b) Consider the sequence $z_n = (1, \dots, \frac{1}{n}, 0, 0, \dots)$. Then, $z_n \to z \in \ell^\infty$, where z is the sequence with $(z)_n = \frac{1}{n}$ for all $n \in \mathbb{N}$. In particular, $z \notin \ell_c$. However, for all $m, n > N$, (assume wlog $m > n$) we have $\|z_n - z_m\|_\infty = \|(0, 0, \dots, 1/(n+1), \dots, 1/m)\|_\infty = 1/(n+1) < 1/N$. Therefore, $\{z_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence, which doesn't converge to any $z' \in \ell_c$.

(c) Suppose $\{z_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in ℓ_0. Since ℓ^∞ is complete, and $\{z_n\}_{n \in \mathbb{N}}$ is also a sequence in ℓ^∞ because $\ell_0 \subset \ell^\infty$, we have that there exists some $z \in \ell^\infty$ such that $z_n \to z$. Let $\epsilon > 0$. Then, there exists some n such that $\|z - z_n\| < \epsilon/2$. Also, there exists some M such that for all $j > M$, $|(z_n)_j| < \epsilon/2$, since $z_n \in \ell_0$. Therefore, for all $j > M$,

$$|(z)_j - (z_n)_j| \leq \sup_j |(z)_j - (z_n)_j| \leq \|z - z_n\|_\infty = \epsilon/2,$$

and so

$$|(z)_j| \leq \epsilon/2 + |(z_n)_j| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since we can find such an M for any $\epsilon > 0$, we conclude that $\lim_{j\to\infty}(z)_j = 0$. Therefore, $z \in \ell_0$, so $(\ell_0, \|\cdot\|_\infty)$ is complete.

(d) We need to show that the limit of every sequence in ℓ_c in ℓ^∞ is in ℓ_0, and that every element of ℓ_0 is the limit of a sequence in ℓ_c.

Suppose $z_n \to z$, where $z_n \in \ell_c$. We show $\lim_{j\to\infty}(z)_j = 0$. Let $\epsilon > 0$. Since $z_n \to z$, there exists some n such that $\|z_n - z\| < \epsilon$. However, since $z_n \in \ell_c$, we have that there exists some M such that $(z_n)_j = 0$ for all $j > M$. But then, for all $j > M$, $|(z)_j - (z_n)_j| = |(z)_j| < \|z_n - z\| < \epsilon$. Since we can findsuch an M for any ϵ, $\lim_{j\to\infty}(z)_j = 0$, so $z \in \ell_0$.

Now, suppose $z \in \ell_0$, so $z = (x_1, x_2, \ldots)$, and define $z_n = (x_1, x_2, \ldots, x_n, 0, 0, \ldots)$. Then, $z - z_n = (0, \ldots, 0, x_{n+1}, x_{n+2}, \ldots)$, so that $\|z - z_n\|_\infty = \sup_{j>n}|(z)_j| \to 0$ since $z \in \ell_0$. Therefore, $z_n \to z$. Thus, ℓ_0 is the smallest closed set which contains ℓ_c.

3. Show that on $C^1([0,1])$:

(a) $\|f\|_0 = |f(0)| + \|f'\|_\infty$ is a norm (where, recall $\|g\|_\infty = \sup_{x\in[0,1]}|g(x)|$);

(b) $\|\cdot\|_0$ is equivalent to the norm $\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty$.

(a) Suppose $\|f\|_0 = 0$. In particular, this implies $\|f'\|_\infty = 0$. Let $x \in (0,1]$. Then, by MVT, there exists some $y \in (0,x)$ such that $f(x) - f(0) = f'(y)(x-0)$. But then, $f'(y) = 0$, so $f(x) = f(0) = 0$. Therefore, $f(x) = 0$.

Next, let $\lambda \in \mathbb{C}$. Then, $(\lambda f)' = \lambda f'$. We compute:

$$\|\lambda f\|_0 = |\lambda f(0)| + \sup_{x\in[0,1]}|\lambda f'(x)| = |\lambda||f(0)| + |\lambda|\sup_{x\in[0,1]}|f'(x)| = |\lambda|\|f\|_0.$$

Finally, by the linearity of the derivative, and the fact that $\|\cdot\|_\infty$ is a norm over continuous function son $[0,1]$ (we checked this in class),we compute the triangle inequality:

$$\begin{aligned}\|f+g\|_0 = |f(0)+g(0)| + \|(f+g)'\|_\infty &\leq |f(0)| + |g(0)| + \|f'+g'\|_\infty \\ &\leq |f(0)| + |g(0)| + \|f'\|_\infty + \|g'\|_\infty \\ &= \|f\|_0 + \|g\|_0.\end{aligned}$$

(b) First, we have

$$\|f\|_0 = |f(0)| + \|f'\|_\infty \leq \|f\|_\infty + \|f'\|_\infty = \|f\|_{C^1}.$$

Next, by the MVT we have that for all $x \in (0,1]$, we have that

$$|f(x)| - |f(0)| \leq |f(x) - f(0)| = |f'(x)|x \leq |f'(x)| \leq \|f'\|_\infty.$$

therefore, $|f(x)| \leq \|f'\|_\infty + |f(0)|$. Since this applies for all $x > 0$, combined with $|f(0)| \leq \|f'\|_\infty + |f(0)|$, we have that $\|f\|_\infty \leq \|f'\| + |f(0)|$ Thus,

$$\|f\|_{C^1} = \|f\|_\infty + \|f'\|_\infty \leq (|f(0)| + \|f'\|_\infty) + \|f'\|_\infty \leq 2(|f(0)| + \|f'\|_\infty) = 2\|f\|_0.$$

Therefore, $\|f\|_0 \leq \|f\|_{C^1} \leq 2\|f\|_0$, which proves these two norms are equivalent.

4. Let X, Y, Z be normed vector spaces. Show:

(a) (leftover from class) the operator norm is a norm on $L(X,Y)$;

(b) if $T \in L(X,Y)$, $S \in L(Y,Z)$, composition $ST \in L(X,Z)$ with $\|ST\| \le \|S\| \, \|T\|$;

(c) for $X = C([0,1])$, $\|\cdot\|_\infty$, find $S, T \in L(X,X)$ with $\|ST\| < \|S\| \, \|T\|$.

(a) Recall that the operator norm is defined as $\|T\| = \sup\{|T(x)|_Y : |x|_X = 1\}$. (I will omit the subscripts, since whether to use the the norm on X or Y is clear from context). Suppose $\|T\| = 0$. If $T \neq 0$, then there exists some $x \in X$ such that $Tx \neq 0$. In that case, $x \neq 0$ since $Tx = 0$ for any linear map T. Then, $0 < |T(x/|x|)| \le \|T\|$, which is a contradiction. Therefore, $T = 0$.
Next, for $\lambda \in \mathbb{K}$,

$$\|\lambda T\| = \sup\{|\lambda T(x)| : |x| = 1\} = \sup\{|\lambda||T(x)| : |x| = 1\} = |\lambda| \sup\{|T(x)| : |x| = 1\} = |\lambda|\|T\|.$$

Finally, for the triangle inequality, let $|x| = 1$. Then,

$$|(T+S)(x)| = |T(x) + S(x)| \le |T(x)| + |S(x)| \le \|T\| + \|S\|.$$

The last inequality follows from the definition of the operator norm because $|x| = 1$. Taking the supremum of $|(T+S)(x)|$ over all such $|x| = 1$, we get that $\|T+S\| \le \|T\| + \|S\|$.

(b) Let $T \in L(X,Y)$ and $S \in L(Y,Z)$. Then, $\|ST\| = \sup\{|TSx| : |x| = 1\}$. But then, for x with $|x| = 1$, we have that either $Tx = 0$ so $STx = 0 \le \|T\|\|S\|$, or $Tx \neq 0$ and

$$|STx| = |Tx||S(Tx/|Tx|)| \le |Tx|\|S\| \le \|T\|\|S\|.$$

Taking the supremum of $|STx|$ over all $|x| = 1$, we get that $\|ST\| \le \|T\|\|S\|$.

(c) For $f \in X$, let $S(f) = f(0)(1-x)$ and let $T(f) = f(1)x$, for $x \in [0,1]$. We show that $T, S \in L(X,X)$. In particular, these are both linear functions, so clearly $S(f), T(f) \in X$. Also, $S(\lambda f + g) = \lambda f(0)(1-x) + g(0)(1-x) = \lambda S(f) + S(g)$, and with a similar calculation for $T(f)$, we show that both S and T are linear maps. Finally, if $\|f\|_\infty \le 1$, then $|T(f)|_\infty, |S(f)|_\infty \le |f(0)|, |f(1)| \le \|f\|_\infty \le 1$, and since this applies for all $\|f\|_\infty = 1$ we have that $\|T\|, \|S\| \le 1$. Therefore, T, F are bounded linear maps so they are in $L(X,X)$.
Now, we compute $\|T\|, \|S\|$. Consider $f(x) = 1$. Then, $\|f\|_\infty = 1$, and $T(f) = x$, $S(f) = 1-x$. We have that $\|T(f)\|_\infty = \|S(f)\|_\infty = 1$. Therefore, $\|T\| = \|S\| = 1$.
However, for all $f \in X$, $ST(f) = (T(f)(0))(1-x) = (f(1)\cdot 0)(1-x) = 0$. Since this applies for all $f \in X$, $ST = 0$, and so $\|ST\| = 0$. But then, $0 = \|ST\| < \|S\|\|T\| = 1 \cdot 1 = 1$.

5. Consider the following spaces of sequences:

$$\ell^1 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \|x\|_1 = \sum_{j=1}^{\infty} |x_j| < \infty\} \qquad \text{(absolutely summable sequences)}$$

$$\ell_0 = \{x = (x_1, x_2, \dots) \mid x_j \in \mathbb{C},\ \lim_{j\to\infty} x_j = 0\} \qquad \text{(sequences tending to 0)}$$

(both are vector spaces). Show that

(a) $\|\cdot\|_1$ is a norm on ℓ^1;

(b) for $y = (y_1, y_2, \dots) \in \ell^1$, the map $f_y : \ell_0 \to \mathbb{C}$ given by $x = (x_1, x_2, \dots) \mapsto y \cdot x = \sum_{j=1}^{\infty} y_j x_j$ is a bounded linear functional on ℓ_0, $\|\cdot\|_\infty$ (i.e. $f_y \in (\ell_0)^*$);

(c) $\|f_y\| = \|y\|_1$;

(d) if $f \in (\ell_0)^*$, then $f = f_y$ for some $y \in \ell^1$;

(e) the map $y \mapsto f_y$ is an isometric isomorphism $\ell^1 \to (\ell_0)^*$.

(a) Let $y \in \ell^1$ and suppose $\|y\|_1 = 0$. Then, for all $j \in \mathbb{N}$ $|y_j| \le \sum_{k=1}^{\infty} |y_k| = \|y\|_1 = 0$. Therefore, $y = 0$. Next, let $\lambda \in \mathbb{C}$, and $y \in \ell^1$. Then,

$$\|\lambda y\|_1 = \sum_{j=1}^{\infty} |\lambda y_j| = \lim_{n\to\infty} \sum_{j=1}^{n} |\lambda y_j| = \lim_{n\to\infty} |\lambda| \sum_{j=1}^{n} |y_j| = |\lambda| \lim_{n\to\infty} \sum_{j=1}^{n} |y_j| = |\lambda| \|y\|_1.$$

Finally, since $y, z \in \ell^1$, both are absolutely convergent, so

$$\|y+z\|_1 = \lim_{n\to\infty} \sum_{j=1}^{n} |y_j+z_j| \le \lim_{n\to\infty} \sum_{j=1}^{n} (|y_j|+|z_j|) = \lim_{n\to\infty} \sum_{j=1}^{n} |y_j| + \lim_{n\to\infty} \sum_{j=1}^{n} |z_j| = \|y\|_1 + \|z\|_1.$$

(b) First, we show that f_y is a linear functional. Let $x \in \ell_0$. Then, there exists some N such that for all $j > N$, $|x_j| \le 1$.

$$\sum_{j=1}^{\infty} |y_j x_j| \le \sum_{j=1}^{N} |y_j x_j| + \sum_{j=N+1}^{\infty} |x_j y_j| \le \sum_{j=1}^{N} |y_j x_j| + \|y\|_1 < \infty.$$

This shows that any $x \in \ell_0$ such sums are absolutely summable. Therefore, since for any $\lambda \in \mathbb{C}$, $x, z \in \ell_0$, the fact that $x + \lambda z \in \ell_0$ and absolute summability gives us that we can break the sum apart:

$$f_y(x + \lambda z) = \sum_{j=1}^{\infty} (x_j + \lambda z_j) y_j = \sum_{j=1}^{\infty} x_j y_j + \lambda \sum_{j=1}^{\infty} z_j y_j = f_y(x) + \lambda f_y(z).$$

Therefore, f_y is a linear functional. To show that f_y is bounded, for any $x \in \ell_0$ with $\|x\|_\infty = 1$, we have that $|x_j| \le 1$, so

$$|f_y(x)| = \left| \sum_{j=1}^{\infty} x_j y_j \right| \le \sum_{j=1}^{\infty} |x_j y_j| \le \sum_{j=1}^{\infty} |y_j| = \|y\|_1.$$

In particular, by the definition of the operator norm, this tells us that $\|f_y\| \le \|y\|_1$.

(c) In part (b), we showed that $\|f_y\| \le \|y\|_1$. Next, define

$$x_N = \begin{cases} y_j^*/|y_j| & \text{if } y_j \neq 0, j \le N \\ 1 & \text{if } y_j = 0, j \le N \\ 0 & \text{else} \end{cases}.$$

Then, by construction, $\|x_N\|_\infty = 1$,

$$|f_y(x_N)| = \sum_{j=1}^{N} |y_j|.$$

But then, since $\lim_{N\to\infty} |f_y(x_N)| = \|y\|_1$, we have that for all $\epsilon > 0$, there exists some N such that $|f_y(x_N)| > \|y\|_1 - \epsilon$. Therefore, $\|f_y\| = \sup\{|f_y(x)| : \|x\|_\infty\} \ge \|y\|_1$, so we deduce $\|f_y\| = \|y\|_1$.

(d) Let $f \in (\ell_0)^*$. Let $e_j \in \ell_0$ be $(0, \dots, 0, 1, 0, \dots)$ where the 1 is in the j-th spot. Now, let y be such that $y_j = f(e_j)$. We show that $y \in \ell^1$.

Define x_N as in part (b), with y. Then, $x_N \in \ell_0$. However, since f is bounded, and $x_N = \sum_{j=1}^N x_j e_j$

$$f(x_N) = \sum_{j=1}^N f(x_j e_j) = \sum_{j=1}^N x_j f(e_j) = \sum_{j=1}^N |y_j|.$$

Now, since f is in $(\ell_0)^*$, we have that f is bounded, and because $\|x_N\|_\infty = 1$, we have that $|f(x_N)| \leq \|f\|$. Then,

$$\lim_{N\to\infty} |f(x_N)| = \lim_{N\to\infty} \sum_{j=1}^N |y_j| = \sum_{j=1}^\infty |y_j| \leq \|f\|.$$

Therefoe, y is absolutely summable, so $y \in \ell^1$. Then, as we saw earlier, for $x \in \ell_0$

$$f_y(x) = \lim_{N\to\infty} \sum_{j=1}^N x_j f(e_j) = \lim_{N\to\infty} \sum_{j=1}^N f(x_j e_j) = \lim_{N\to\infty} f\left(\sum_{j=1}^N x_j e_j\right) = f(x),$$

as required, where in the last step we use the continuity of f, and the fact that $\| \sum_{j=1}^N x_j e_j - x\|_\infty \to 0$ as $N \to \infty$, because $x_j \to 0$ since $x \in \ell^0$.

(e) We have seen already that $y \mapsto f_y$ is an injective map from $\ell^1 \to (\ell_0)^*$:

- Map: part (b) shows that f_y is a linear function on ℓ_0.
- Injective: part (c) shows that if $y_1 \neq y_2$, then $\|y_1 - y_2\|_1 \neq 0$, and then $\|f_{y_1 - y_2}\| = \|y_1 - y_2\|_1 \neq 0$ so $f_{y_1 - y_2} \neq 0$.
- Surjective: part (d) shows that if $f \in (\ell_0)^*$, there exists $y \in \ell^1$ such that $f = f_y$.
- Isometry: $\|f_y\| = \|y\|_1$.

Therefore, $y \mapsto f_y$ is an isometric isomorphism.

6. On $(\ell^\infty, \|\cdot\|_\infty)$, define the shift operator $S : (x_1, x_2, \dots) \mapsto (x_2, x_3, \dots)$, the subset $Y = \{x - Sx \mid x \in \ell^\infty\} \subset \ell^\infty$, and the sequence $z = (1, 1, 1, \dots) \in \ell^\infty$. Show:

(a) S is linear and Y is a subspace;

(b) $\operatorname{dist}(z, Y) = \inf_{y \in Y} \|z - y\|_\infty = 1$;

(c) by applying (a Corollary of) the Hahn–Banach theorem to $\overline{Y}$ and z, that there is $T \in (\ell^\infty)^*$ with $T\restriction_Y = 0$, $T(z) = 1$ and $\|T\| = 1$;

(d) T is shift-invariant: $TS = T$;

(e) if $x = (x_1, x_2, \dots) \in \ell^\infty$ satisfies $\lim_{k\to\infty} x_k = L$ then $Tx = L$.

(a) To show S is linear, let $x, y \in \ell^\infty$. Then, for $\lambda \in \mathbb{C}$

$$S(x + \lambda y) = (x_2 + \lambda y_2, \dots) = (x_2, \dots) + \lambda(y_2, \dots) = S(x) + \lambda S(y).$$

To show Y is a subspace, we see that $z \in \ell^\infty$ and $0 = z - Sz$, so $0 \in Y$. Next, for $x, y \in Y$, we have some $x', y' \in \ell^\infty$ such that $x = x' - Sx'$ and $y = y' - Sy'$. Then, for $\lambda \in \mathbb{C}$, we have by the linearity of S that

$$x + \lambda y = x' - Sx' + \lambda y' - S\lambda y' = (x' + \lambda y') - S(x' + \lambda y').$$

Therefore, $x + \lambda y \in Y$.

(b) To see that $\inf \|z-y\|_\infty \leq 1$, we notice that $0 \in Y$ since Y is a subspace, and $\|z-0\|_\infty = 1$. For the other direction, suppose for the sake of contradiction that there exists $y \in Y$ such that $\|z-y\|_\infty < 1$. Then, $\|z-y\|_\infty < 1-\epsilon$ for some $\epsilon > 0$. In particular, for each $j \in \mathbb{N}$, $|z_j - y_j| = |1-y_j| < 1-\epsilon$, so $\epsilon < y_j < 2-\epsilon$. However, there exists some $x \in \ell^\infty$ such that $y = x - Sx$, so that $y_j = x_j - x_{j+1}$. Then, we notice that $\sum_{j=1}^N y_j = x_1 - x_{N+1}$. However, $N\epsilon < \sum_{j=1}^N y_j = x_1 - x_{N+1}$, which implies that $N\epsilon - |x_1| \leq |x_{N+1}|$. Therefore, $|x_{N+1}| \to \infty$ as $N \to \infty$, which contradicts that $x \in \ell^\infty$ so that $|x_N| \leq \|x\|_\infty$ for all $N \in \mathbb{N}$. Therefore, it cannot be the case that $\|z-y\|_\infty < 1$, so we must have $\inf_{y \in Y} \|z-y\|_\infty \geq 1$.

(c) Since $\|z-y\| \geq 1$ for all $y \in Y$, we have that z is not a limit point of any sequence of points in Y, so $z \notin \overline{Y}$. Therefore, $z \in \ell^\infty \setminus \overline{Y}$, and we can apply Theorem 5.8 a. from Folland (a Corollary of the Hahn Banach theorem) to produce a linear map $T \in (\ell^\infty)^*$ with $T(z) = 1 = \inf_{y \in \overline{Y}} \|y-z\|$ and $\|T\| = 1$. (In particular, $\inf_{y \in \overline{Y}} \|y-z\| = \inf_{y \in Y} \|y-z\|$) because the norm map is a continuous map, and any limit point is arbitrarily close to points in Y).

(d) Let $x \in \ell^\infty$. Let $y \in Sx - x$, so $y \in Y$. Then,

$$(TS-T)x = TSx - Tx = T(Sx-x) = Ty = 0,$$

since $T\restriction_Y = 0$. However, this implies that $TSx = Tx$. Since x was arbitrary, we have that $TS = T$, as desired.

(e) Since T is shift invariant, we have that for all $n \in \mathbb{N}$, $TS^n = T$. Let $\epsilon > 0$. We will show that $Tx = T(L \cdot z) = LT(z) = L$. Since $\lim_{k \to \infty} x_k = L$, we know that there exists some N such that for $k \geq N$, $|x_k - L| < \epsilon$. Then, by shift invariance of T and z, we have that

$$T(x - L \cdot z) = TS^N(x - L \cdot z) = T(S^N x - L \cdot S^N z) = T(S^N x - L \cdot z).$$

However, $S^N x - L \cdot z = (x_N - L, x_{N+1} - L, \ldots)$ so we know that $\|S^N x - L \cdot z\| < \epsilon$. This then implies that

$$|T(x - L \cdot z)| \leq \|T\| \|S^N x - L \cdot z\| < 1 \cdot \epsilon.$$

Since ϵ is arbitrary, it follows that $|T(x - L \cdot z)| = 0$, so $Tx = LT(z) = L$, as desired.